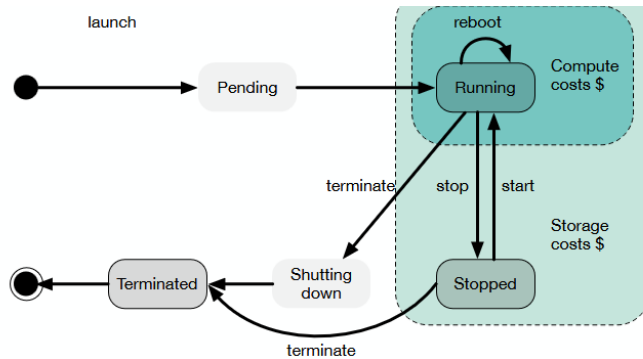
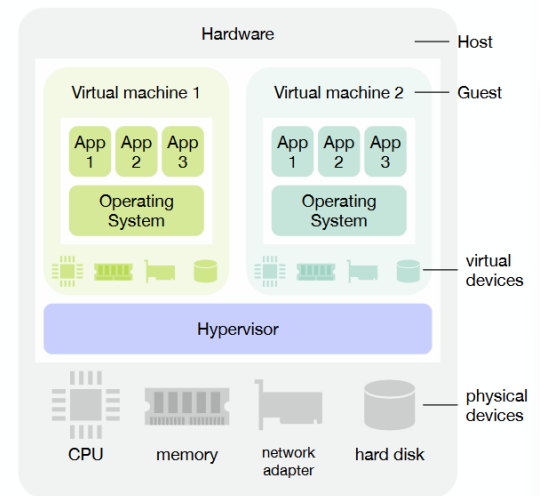


Ch1 base

Instance : {Where, os, CPU cores, memo, disk size, rules of firewall, authentication}



Server with virtualization (type 1 hypervisor)



Ch2 concepts overview

- **Cloud computing** : A cloud provider provides the infrastructure as a service to a cloud client which uses it to provide applications and services to its users.
 - **AWS** : Cloud computing is the on-demand delivery of compute power, database, storage, applications, and other IT resources via the internet with pay-as-you-go pricing.
 - a. **On-demand self serviced** Automatic provisioning without requiring human interaction
 - b. **Broad network access** : Access via standardized protocols from a variety of clients
 - c. **Resource pooling** : Serve multiple customers in a multi-tenant model, dynamic assignment of resources from a pool, location independence
 - d. **Rapid elasticity** : Rapid provisioning/deprovisioning to scale out/in, seemingly unlimited capacity
 - e. **Measured service** : Usage is monitored and controlled, providing transparency in billing
 - **Data security** : trust, en gros c'est comme les bank (tk)
 - 3 rules of trust
 - a. All things being equal, minimize the number of organizations and people whom you have to trust.
 - b. Use evidence and experience to measure trustworthiness.
 - c. Trust proportionally to risk.
 - **SaaS (Software as a Service)** = app dans le browser
 - **PaaS (Platform as a Service)** = hardware/software plftaore for dev create cloud-base applications
 - En gros, c'est un ensemble de tool qui permette de faire des app sur le provider
 - **IaaS (Infrastructure as a Service)** = Basic IT ressources in virtual form
 - virtual machines
- **Public cloud** : tous le monde l'utilise, y a un system de credit
 - AWS, Azure, OVH
- **Private cloud** : cloud d'entreprise
- **Community cloud** : tous petit cloud
- **Hybird cloud** : utiliser plusieurs type de cloud

Ch3 Infrasctrures as a service

Open source : Openstack

- hypervisor : crée et manage les VM
- EC2 Instance : virtsual machine (AWS)
- USAGE : Charge base on time using the VM
- Stokage : Charge by capacity * time

- On-Demande Instance : pay by hours
- Reserved Instance : Pay yearly upfront
- Spot Instance : use spare Instance for cheaper

APP state probleme

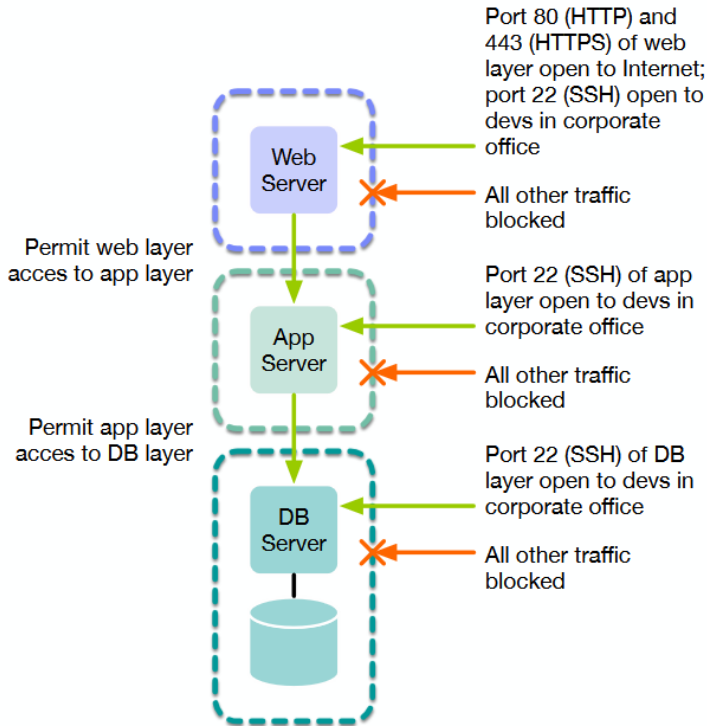
- Stateless = no data need to respond
- Stateful = need data to respond (e-commerce apps)
- Soft state = data is need but is not permanante (electronic shopping cart ?)

Pricing

Charge of data transfer

En gros, on fait payer souvent que quand on sort des donnée pour eviter que les gens se case

Virtual machine	charge per time running
Operating system	Open Source: free, licensed OS: charge per time running
Virtual disk	charge per storage capacity and time allocated
Public/private key pair	no charge
VM firewall	no charge
VM firewall rules	no charge
Public IPv4 address	charge per time allocated
Static public IPv4v address	charge per time under customer's control
Monitoring of VM	no charge
Data transfer	charge per amount of data transferred out of the cloud



Ch4 Scaling apps on laaS

Load balancer peut marcher dans plusieurs niveau

- TCP (layer 4) (**FAST**) = directly open TCP connection with server -> good
- HTTP (layer 7) (**SLOW**)(**SMARTER**) = decode the HTTP request -> decide where to forward it -> good

Cloud load balancer (IaaS)

Pricing = up time * volume

Network load balancer = TCP

App load balancer = HTTP (3 part)

- Load balancer
 - Listener = listen HTTP request
 - Target groups = groups of VM to be forwarded
- VPC = Virtual Private Cloud = Collection of virtual network in a region

Tier = physical separation, layer = logical

Cloud computing

Not to be confused with Cloud Computing (horse).

